W10 - *PRACTICE*

Monster Slayer

At the end of this practice, you should be able to…

- ✔ Use all concepts you have learned so far to create a complex project

How to work?

- ✔ Download **the start code** from the Google classroom
- ✔ For each exercise you can either:
  - o Run `npm install`
  - o Or move an existing `node_modules` to the exercise folder *(fastest option!)*

How to submit?

- ✔ **Create a repository on GitHub** with the name of this practice:
  Ex: C2-S1-PRACTICE
- ✔ **Push your final code** on this GitHub repository (if you are lost, [follow this tutorial])
- ✔ Finally, submit on **Google classroom** your GitHub repository URL
  Ex: `https://github.com/thebest/` C2-S1-PRACTICE`.git`

Are you lost?

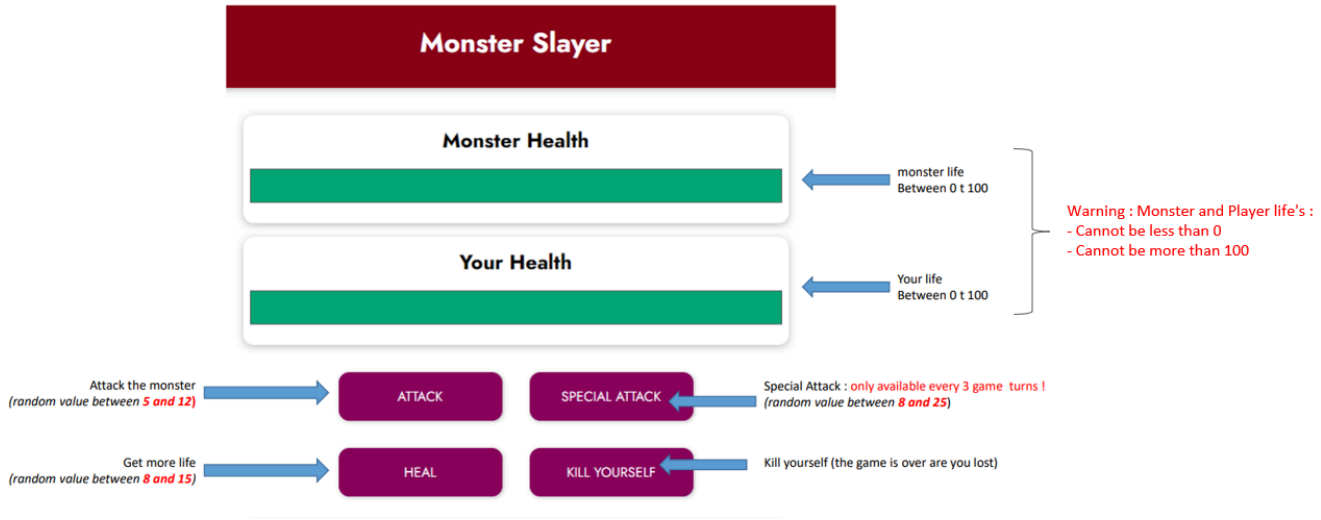
You can read the following documentation to be ready for this practice:
https://www.w3schools.com/react/react_jsx.asp
https://www.w3schools.com/react/react_props.asp
https://www.gatsbyjs.com/docs/how-to/images-and-media/importing-assets-into-files/

Understand the game rules

Start with playing the game online, and understand its rules:
https://reactjs-game-practice.vercel.app/

In this game you need to fight against a monster:

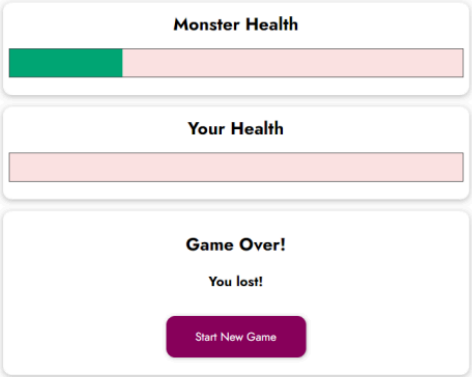
- You can choose to **attack** *(the monster will attack back)*
- Or **heal** yourself *(the monster will also attack back)*
- Every 3 turns, you can use a **special attack**
- You can also **kill** yourself if you want 😊

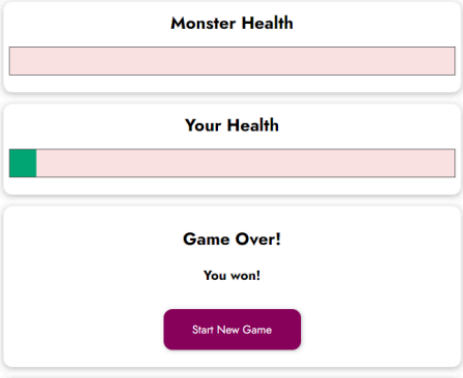


The game will also contain a dedicated component to display all actions:

Finally, the game over component will display the result (*you won, you lost, draw game*).

- The player can also **start a new game**





1. Vanilla HTML to React components

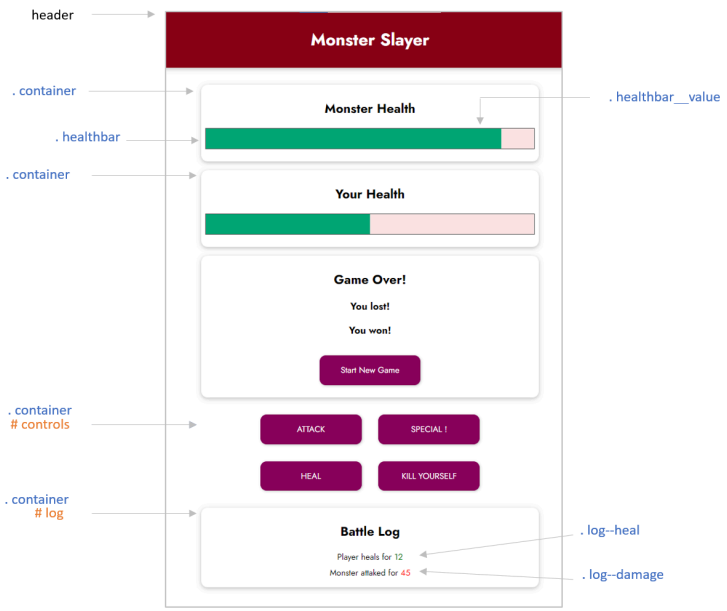
As start code we provide:

- A **vanilla HTML CSS** project, *which serves as example to style and structure your HTML*
- A **start REACT JS** project, *that will contain your code*

The goal of this first activity is only to **move** the HTML and CSS code from vanilla project to your new React JS components….

Q1 – First understand the vanilla HTML CSS project:

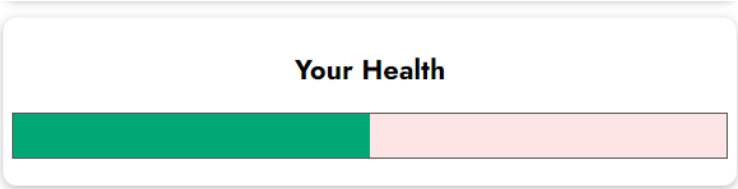
- *Which CLASSES are used?*
- *Which ID are used?*



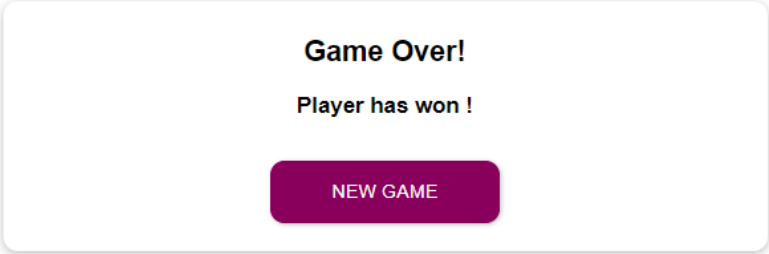
Q2 – Then, In your REACT JS project, create the components and move the HTML/CSS code

- **Header**: *representing the game header*

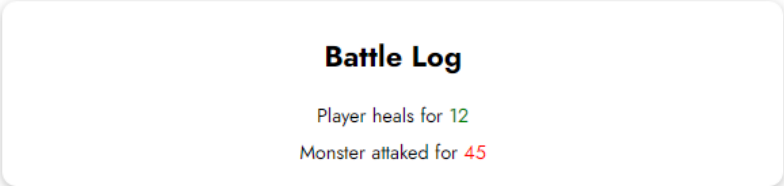
- **Game**: *containing all game logic and game sub components*
- **Entity**: *representing the monster or player's heath*



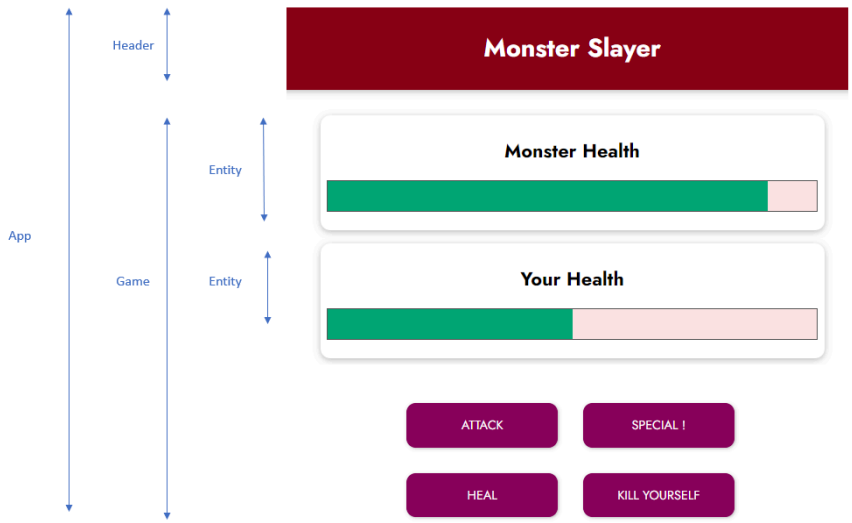
- **Game Over**: *informing about the end of the game:*



- **Log**: *representing all the game actions*



<u>As a result, here is how your app components and its sub component could look like</u>



Q3 – Draw the **component diagram** (power point) to represent how your components communicate.

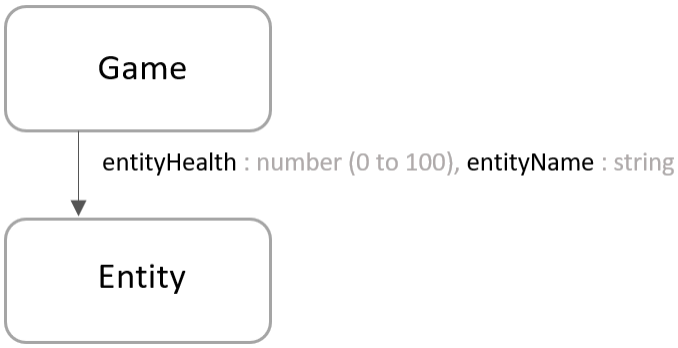
2. Entity component

The **Entity** component gets 2 parameters:

- The health percentage *a number from 0 to 100*
- The entity name: either Player or Monster

Q1 – Complete your code

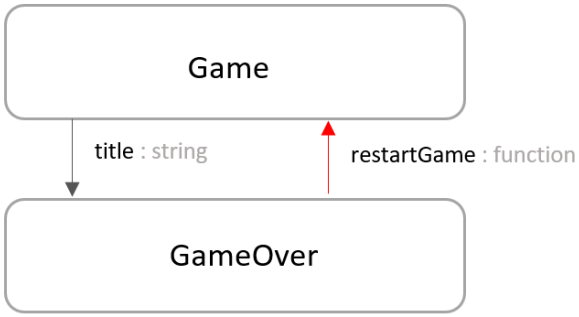
- To display the health bar properly in the Entity component
- To create 2 entities (monster and player) in the Game component



3. Game over component

The **GameOver** component gets 2 parameters:

- The title: *the game over message to display in the <h3>*
- The restartGame: *a function reference to be called when user click on start new game.*



4. Log component

Each log message can be stored using the following data structure:

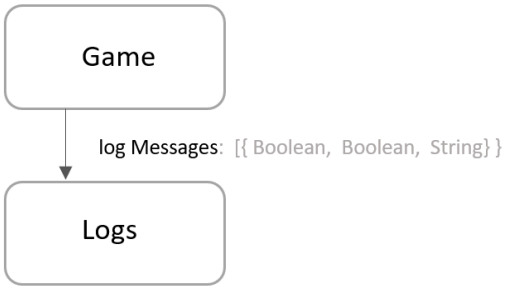
- **isPlayer**: whether its s the player or the monster
- **isDammage**: whether it's a damage or a heal
- **text**: the text to display

Log
isPlayer: Boolean
isDamage: Boolean
text: String

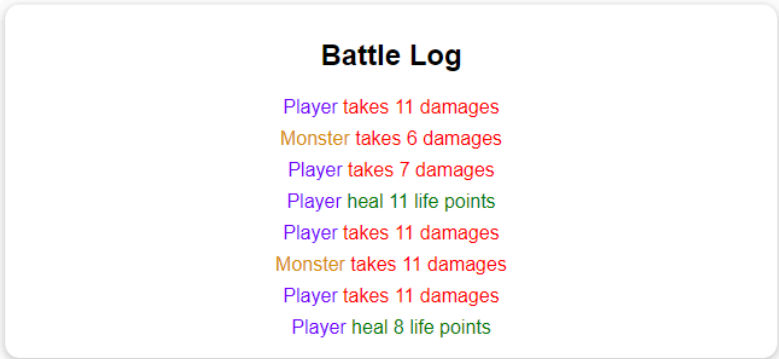
<u>Note</u>: we already have created some helper function in the Game component to create log messages.

The **Logs** component gets 1 parameter:

- The list of logs, following the above structure



The **Logs** component shall display each log, following the color specification:



5. Game component

The **Game** component handles the game logic. It will be the **main component** of your project.

Q1 – Take time to list down the *states* and the *computed values* you will need to handle all your game features.

Note: Computed values are data that do not need a state, as they can be computed from other data (state, constant, variable.)

For example: how will you manage whether the game is over, who has won? Whether the special attack need to be displayed or not? How will you store the logs? Etc.

State	Type	Role
playerHealth	number	Update player's health
monsterHealth	number	Update monster's health
logs	array	Store game history

Computed values	Type	Role

isGameOver	boolean	Determine if game is over
hasPlayerWon	boolean	Determine if player won

Q2 – Link your states ad computed value to your component's attributes

- Visibility of the GameOver, the buttons
- The monster and player health bar
- The log component

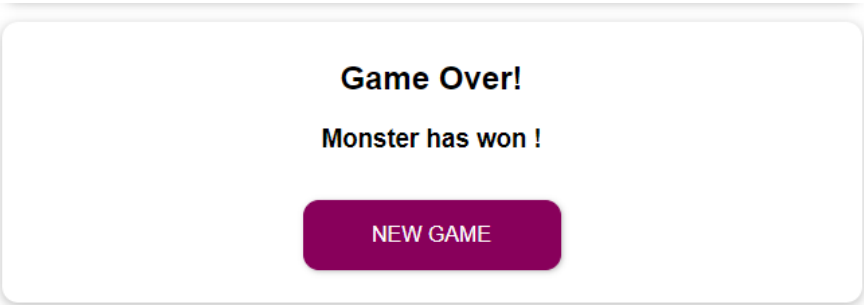
Q3 – Handle **actions** related to each button, following the **specifications**.

Note: we already have created a helper function in the Game component to generate a random value between a min and a max value.

Q4 – Handle **the end of the game**

The GameOver component shall display when the game is over, and propose the player to start a new game

Note: What do you need to update to start a new game?



6. Test and deploy on Vercel

Q1 – You game should handle **every situation**

For instance, did you think about those difference cases?

- What s happen in case of draw game?
- Can player or monster have more than 100% life?

- Can player or monster have less than 0% life?

Q2 – Before deploying, clean up your code

- Each function needs comments
- Remove useless code, useless variables…

Q3 – Push your code on GitHub and deploy on Vercel (search on internet how to deploy)